UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #3

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
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Your name:____
Due Date: Wednesday February 19 by 5:00pm in my office

Number of points...............................100
Your score..................................

TABLE #1

Symbol	Definition, Units
T_R	Reactor Temperature, C
V_R	Reactor volume, m^3
C_{AA}	Acetic acid molarity, $kmol/m^3$
C_{ETOH}	Ethanol molarity, $kmol/m^3$
C_{EAC}	Ethyl acetate molarity, $kmol/m^3$
C_{WATER}	Water molarity, $kmol/m^3$
k_{fo}	Pre-exponential factor, $m^3/(kmol \cdot s)$
k_{ro}	Pre-exponential factor, $m^3/(kmol \cdot s)$
E_{fo}	Activation energy, kJ/mol
E_{ro}	Activation energy, kJ/mol
Symbol	**Data for Problem 1B**
k_{fo}	$1.9\times10^8\ m^3/(kmol \cdot s)$
k_{ro}	$5.0\times10^7\ m^3/(kmol \cdot s)$
E_{fo}	59.50 kJ/mol
E_{ro}	59.50 kJ/mol

Note: **All problems are worth 100 points. You final score will be the average value**

Problem 3A: Calculation of Process Gains for Optimal Controller Tuning

The esterification of acetic acid with ethanol takes place in a water cooled jacketed Continuous Stirred Tank Reactor (CSTR) under atmospheric conditions. The reactor volume is $\{V_R = 10\, m^3\}$ and the reaction mechanism and kinetic constants are given in the table below:

Reaction	$CH_3COOH + C_2H_5OH \underset{k_r}{\overset{k_f}{\rightleftharpoons}} CH_3COOC_2H_5 + H_2O$
Reaction Rate	$\Re_1 = k_f C_{AA} C_{ETOH} - k_r C_{EAC} C_{WATER}$
Forward Rate Constant	$k_f = k_{f0} \exp\left(-\frac{E_f}{R_g T_R}\right)$
Reverse Rate Constant	$k_r = k_{r0} \exp\left(-\frac{E_r}{R_g T_R}\right)$

The conditions of the feed to the reactor are given in the table below:

Feed	Data
Temperature	$25^o C$
Pressure	$1\, atm$
Volumetric Flow Rate	$120\, gpm$
Acetic Acid Mole Fraction	0.495
Ethanol Mole Fraction	0.479
Ethyl Acetate Mole Fraction	0.000
Water Mole Fraction	0.026

The cooling water is available at $\{25^o C\}$ and a flow rate of $\{97\, gpm\}$. The residence time of the water in the cooling jacket is approximately one tenth of the residence time in the reactor. To control the temperature of the reactor we are considering traditional feedback control by manipulating the cooling water flow rate, feedback combined with cascade control and feedback combined with feed forward control. All modes of the proposed control schemes are shown in the schematic below. For controlling the cooling water flow rate we are considering a 3 inch equal percentage control valve. The valve pressure drop is a weak function of flow rate so it can be considered constant and the valve coefficient as a function of stem position(as a percentage of total travel) is given by the following data:

x(%)	0	10	20	30	40	50	60	70	80	90	100
CV(x)	0	3.11	5.77	9.12	13.7	21.7	36	60.4	86.4	104	114

At steady-state conditions the valve is 60% open. In order to tune the proposed controller(s) we must calculate the process gains for the control various configurations. This can be accomplished by performing appropriate simulation tests. All tests should include a step up, a step down and then a step back to the steady state-value. All tests must be shown graphically along with the detailed calculation of the process gain.

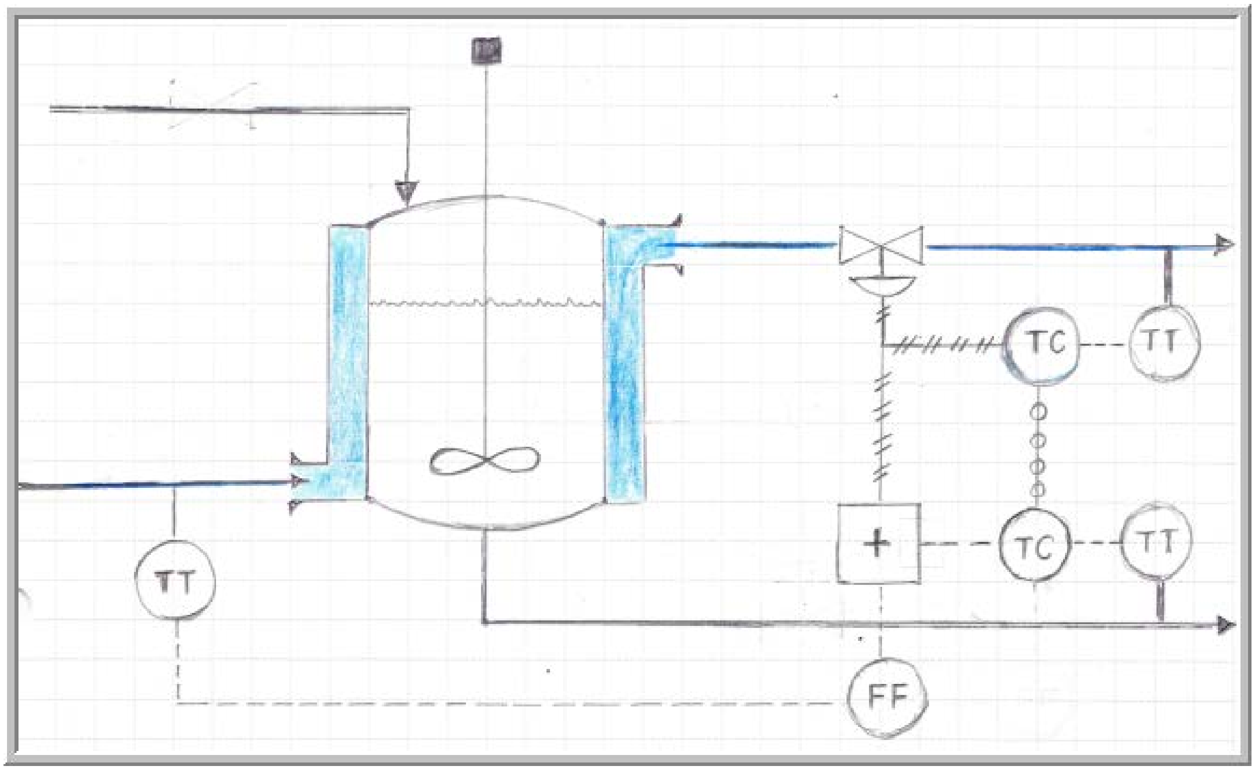

Figure 1. Various Control Schemes for CSTR Temperature Control

Your final report must include the following:

1. The set points for all controllers
2. The process gain for feedback controller

$$\left\{ K_{P,FB} \, [=] \, \frac{\Delta^{o}C\left(\text{Reactor Temperature}\right)}{\%\Delta\left(CO\right)} \right\}$$

3. The process gains for the cascade controller:

$$\left\{ K_{P1,CS} \, [=] \, \frac{\Delta^{o}C\left(\text{Cooling Jacket Temperature}\right)}{\%\Delta\left(CO\right)} \right\}$$

$$\left\{ K_{P2,CS} \, [=] \, \frac{\Delta^{o}C\left(\text{Reactor Temperature}\right)}{\Delta^{o}C\left(\text{Cascade Set Point Temperature}\right)} \right\}$$

4. The process gain for the feed forward controller

$$\left\{ K_{P,FF} \, [=] \, \frac{\Delta^{o}C\left(\text{Reactor Temperature}\right)}{\Delta^{o}C\left(\text{Cooling Water Inlet Temperature}\right)} \right\}$$

5. All graphs indicating the stepped process variable and the corresponding response
6. The Athena(Matlab or Excel) code for parts 2, 3 and 4